

Aldehydes, Ketones and Carboxylic Acids

NCERT Class XII Chemistry · Chapter 8 · Aldehydes, Ketones and Carboxylic Acids

Aldehydes, Ketones and Carboxylic Acids — Complete Notes

8 marks in the board paper (CBSE 2026-27 curriculum), the heaviest of the four organic chapters you are adding, and the one with the most reactions. The whole chapter hangs on one idea: the carbon–oxygen double bond, $C=O$. In aldehydes and ketones that carbon is eager to be attacked, and most of the chapter is "what attacks it, and what do you get". In carboxylic acids the same carbon is less eager, and the interest moves to the acidic $O-H$. Every new term is tagged [exposure] and explained from scratch.

The carbonyl family — one functional group, six families — and why they smell nice

[exposure] The **carbonyl group** is a carbon doubly bonded to oxygen, $C=O$. A **functional group** is a small unit of atoms that decides how a whole organic compound behaves, so every compound that carries it reacts in a family way. The carbonyl group is the centre of this chapter, and these are the families it defines:

Family	General formula	Example
Aldehyde	$R-CHO$	CH_3CHO
Ketone	$R-CO-R'$	CH_3COCH_3
Carboxylic acid	$R-COOH$	CH_3COOH
Acid (acyl) halide	$R-COX$	CH_3COCl
Acid anhydride	$R-CO-O-CO-R$	ethanoic anhydride
Ester	$R-COOR'$	ethyl ethanoate
Amide	$R-CONH_2$	ethanamide

The last four are **derivatives** of carboxylic acids: each is an acid with its $-OH$ swapped for something else. An **anhydride** forms when two acid molecules lose one water between them.

Aldehydes, ketones and carboxylic acids are widespread in plants and animals and play a part in life's biochemistry. Many are responsible for fragrance and flavour: **vanillin** (from vanilla beans), **salicylaldehyde** (from meadowsweet) and **cinnamaldehyde** (from cinnamon).

Naming aldehydes — common names come from the acid; IUPAC drops the "e" and adds "-al"

Common names

An aldehyde's common name comes from the common name of the matching carboxylic acid: replace **-ic acid** with **-aldehyde**. $HCOOH$ is formic acid, so $HCHO$ is **formaldehyde**; acetic acid gives **acetaldehyde** (CH_3CHO); propionic acid gives propionaldehyde; butyric acid gives **butyraldehyde**. Substituents are

located with Greek letters: the carbon next to -CHO is α , then β , γ , δ . So $\text{CH}_3\text{CH}(\text{Br})\text{CH}_2\text{CHO}$ is β -bromobutyraldehyde.

IUPAC names

1. Take the **longest chain that includes the -CHO carbon**. The aldehyde carbon is always **carbon 1**, so it needs no locant.
2. Drop the final "e" of the alkane and add **-al**: CH_3CHO = **ethanal**; $\text{CH}_3\text{CH}_2\text{CHO}$ = **propanal**; $\text{CH}_3\text{CH}_2\text{CH}_2\text{CHO}$ = **butanal**. $\text{CH}_3\text{CH}(\text{Br})\text{CH}_2\text{CHO}$ = **3-bromobutanal**.
3. **Aldehyde on a ring**: name the ring and add **-carbaldehyde**: cyclopentanecarbaldehyde; benzenecarbaldehyde (= **benzaldehyde**, which IUPAC also accepts).
4. **Aldehyde on a side chain of a ring**: the chain is the parent and the ring is a substituent: $\text{C}_6\text{H}_5\text{CH}_2\text{CH}_2\text{CHO}$ is 3-phenylpropanal.
5. **Three or more -CHO groups**: they cannot be counted as chain carbons. Number only the backbone and call them **carbaldehydes**: propane-1,2,3-tricarbaldehyde.

Naming ketones, and which group wins — *the ketone is "-one"; as a prefix it becomes "oxo"*

Common names

Name the two groups on the carbonyl carbon alphabetically, then write **ketone**: CH_3COCH_3 is dimethyl ketone (and **acetone**, its historical name); $\text{CH}_3\text{COC}_2\text{H}_5$ is **ethyl methyl ketone**. Substituents are located with α , α' , β , β' . A ketone with a phenyl group is named with an **acyl** prefix and **-phenone**: $\text{C}_6\text{H}_5\text{COCH}_3$ is **acetophenone**, $\text{C}_6\text{H}_5\text{COC}_6\text{H}_5$ is **benzophenone**.

IUPAC names

1. Longest chain containing the carbonyl; number it so the carbonyl gets the **lowest number**.
2. Drop the "e" and add **-one**: CH_3COCH_3 = **propanone** (propan-2-one); pentan-3-one; cyclopentanone.
3. Substituents are prefixes with locants: 2,4-dimethylpentan-3-one.

[exposure] When a compound carries **two functional groups**, only one can be the "suffix" (the ending of the name). The other has to become a **prefix**. The ranking that matters here: **carboxylic acid** > **aldehyde** > **ketone** > **alcohol**. A ketone that loses out becomes the prefix **oxo-**; an alcohol that loses out becomes **hydroxy-**. Example: a chain with a -CHO at one end and a ketone at C3 is **3-oxopentanal**, and a chain with a ketone and an alcohol is **4-hydroxypentan-2-one** (2014 Q1).

Examiner asks: "write the IUPAC name" for a two-group compound (2014 Q1), and "draw the structure of 4-methylpent-3-en-2-one" or "p-methylbenzaldehyde" (2015 Q20). NCERT intext 8.1 asks for structures from names; the pattern is always chain first, then group positions.

Structure of the carbonyl group — *flat, polar, and electron-poor at carbon — the cause of everything that follows*

The carbonyl carbon is sp^2 **hybridised**. It makes three sigma bonds (one to oxygen, two to R or H) lying in **one plane at about 120°** . Its leftover unhybridised p orbital overlaps sideways with a p orbital on oxygen to form the **pi bond**, above and below the plane.

Oxygen is more electronegative than carbon, so the pi electrons sit nearer oxygen and the bond is **polar**: $\overset{\delta+}{\text{C}} = \overset{\delta-}{\text{O}}$. Resonance puts the same picture another way — one contributor has a positive carbon and a negative oxygen.

[exposure] An **electrophile** is an electron-seeking species and a **nucleophile** is an electron-rich one that attacks a positive centre (you met the nucleophile in Haloalkanes). So the **carbonyl carbon is an electrophilic centre** (a Lewis acid) and the **carbonyl oxygen is a nucleophilic centre** (a Lewis base). Because of the strong dipole, aldehydes and ketones are more polar than ethers.

Preparing aldehydes and ketones — *first the methods that give either; then the aldehyde-only and ketone-only routes*

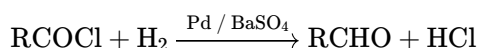
Methods that give aldehydes or ketones

1. **Oxidation of alcohols.** A **primary** alcohol gives an aldehyde, a **secondary** alcohol gives a ketone — with a *mild* oxidant that stops there. **[exposure]** PCC (pyridinium chlorochromate) and anhydrous CrO_3 are the two mild chromium oxidants; PCC is preferred for primary alcohols because it stops at the aldehyde and leaves a $\text{C}=\text{C}$ alone. A strong oxidant would carry an aldehyde on to the acid.
2. **Dehydrogenation of alcohols.** Alcohol vapour passed over copper (or silver) at **573 K** loses hydrogen: primary $\rightarrow$ aldehyde, secondary $\rightarrow$ ketone. It is an industrial method.
3. **Ozonolysis of alkenes.** O_3 and then $\text{Zn} / \text{H}_2\text{O}$ cut the double bond and put $\text{C}=\text{O}$ on each end. Propene gives ethanal and methanal; 2-methylbut-2-ene gives ethanal and propanone (an aldehyde and a ketone).
4. **Hydration of alkynes.** Water adds to a triple bond with HgSO_4 and dilute H_2SO_4 , following Markovnikov. The first product, an enol, flips to the carbonyl form (**tautomerism**). Ethyne gives **ethanal**; every other alkyne gives a **ketone** (propyne gives propanone).

Methods for aldehydes only

Rosenmund reduction — an acid chloride, hydrogenated over **palladium on barium sulphate**.

[exposure] The BaSO_4 **poisons** the palladium (partly deactivates it) on purpose, so it stops at the aldehyde instead of carrying on to the alcohol.



Stephen reaction — a nitrile is reduced to an **imine** $\text{RCH}=\text{NH}$ with stannous chloride and HCl ; hydrolysis gives the aldehyde. **DIBAL-H** (diisobutylaluminium hydride) does the same on nitriles *and* esters, and it does not touch a $\text{C}=\text{C}$.



Oxidising a ring methyl to an aldehyde (toluene $\rightarrow$ benzaldehyde). Strong oxidants overshoot to benzoic acid, so a reagent is needed that stops at an intermediate. Three ways:

- **Etard reaction:** chromyl chloride CrO_2Cl_2 in CS_2 gives a chromium complex; hydrolysis gives benzaldehyde.
- CrO_3 in **acetic anhydride** (273–283 K) gives **benzylidene diacetate**, which hydrolyses to benzaldehyde.

- **Side-chain chlorination:** Cl_2 in light (free-radical) puts two chlorines on the methyl to give **benzal chloride** $\text{C}_6\text{H}_5\text{CHCl}_2$; hydrolysis at 373 K gives benzaldehyde. This is the commercial method.
- 1. **Gattermann–Koch reaction** — benzene with CO and HCl over anhydrous AlCl_3 (and CuCl) gives benzaldehyde. $\text{CO} + \text{HCl}$ gives HCOCl ; AlCl_3 pulls off Cl^- to leave the **formyl cation** HCO^+ , the electrophile that attacks the ring.

Methods for ketones only

1. **From an acid chloride and dialkylcadmium.** A Grignard reagent with CdCl_2 gives R_2Cd ; an acyl chloride then transfers one R to give a ketone. $2 \text{CH}_3\text{COCl} + (\text{CH}_3)_2\text{Cd} \longrightarrow 2 \text{CH}_3\text{COCH}_3 + \text{CdCl}_2$.
2. **From a nitrile and a Grignard reagent**, then hydrolysis: $\text{CH}_3\text{CN} + \text{CH}_3\text{MgBr}$ gives an adduct that hydrolyses to **propanone**.
3. **Friedel–Crafts acylation** — benzene with an acid chloride over anhydrous AlCl_3 :
 $\text{C}_6\text{H}_6 + \text{CH}_3\text{COCl} \xrightarrow{\text{AlCl}_3} \text{C}_6\text{H}_5\text{COCH}_3 + \text{HCl}$ (acetophenone). The AlCl_3 is a Lewis acid that pulls off Cl^- and leaves the **acylium ion** CH_3CO^+ , which attacks the ring.

Trap: know which reagent belongs to which product. Choosing between oxidants: **PCC** for primary alcohol $\rightarrow$ aldehyde (and with an allylic double bond present, PCC leaves it alone); **anhydrous** CrO_3 for secondary alcohol $\rightarrow$ ketone; CrO_2Cl_2 **then** H_3O^+ for toluene $\rightarrow$ benzaldehyde; O_3 **then** $\text{Zn}/\text{H}_2\text{O}$ for but-2-ene $\rightarrow$ two ethanal. This is NCERT Example 8.1 and it is asked as "name the reagent".

Examiner asks: the Rosenmund (2019 Q9), Stephen and Etard equations (2015 Q22), Friedel–Crafts acylation (Q10), the dialkylcadmium route (2022 Q28(a)), DIBAL-H (2017 Q2), and conversions that use them (2022 Q24(a)).

Physical properties of aldehydes and ketones — *dipole–dipole forces put them between hydrocarbons and alcohols*

- **State.** Methanal is a gas; ethanal is a volatile liquid; the rest are liquids or solids.
- **Boiling point.** Higher than hydrocarbons and ethers of similar mass, because of **dipole–dipole attraction** between the polar $\text{C}=\text{O}$ groups. Lower than alcohols of similar mass, because they have **no O–H** and so cannot hydrogen-bond to each other. NCERT's set (masses 58–60): **n-butane 273 K < methoxyethane 281 K < propanal 322 K < propanone 329 K < propan-1-ol 370 K**. Between the two isomers, the ketone (two R groups, more surface area) boils slightly higher than the aldehyde.
- **Solubility.** The lower members — methanal, ethanal, propanone — mix with water in all proportions, because the carbonyl oxygen hydrogen-bonds to water. Solubility falls quickly as the alkyl chain grows (the chain is hydrophobic). All are fairly soluble in organic solvents.
- **Odour.** Lower aldehydes smell sharp and pungent; as the molecule grows the smell becomes less pungent and more fragrant, so they are used in perfumes and flavourings.

Examiner asks: "arrange in increasing boiling point" — propane < ethanal < ethanol (2022 Q5(b)); NCERT example 8.2 (an aldehyde, an alcohol, an ether and a hydrocarbon of similar mass: hydrocarbon < ether < aldehyde < alcohol).

Nucleophilic addition — the master reaction — *a two-step attack on the carbonyl carbon, and why aldehydes win over ketones*

[exposure] Nucleophilic addition is the characteristic reaction of the carbonyl group: a nucleophile attacks the electron-poor carbon and something else adds to the oxygen, so the $C=O$ becomes a single bond and the carbon turns from sp^2 to sp^3 . It happens in two steps:

1. **Slow step.** The nucleophile, coming at roughly right angles to the flat carbonyl plane, bonds to the carbon. The pi electrons move to oxygen, which becomes negative. This gives a **tetrahedral alkoxide intermediate** (the carbon is now sp^3).
2. **Fast step.** The oxygen picks up H^+ from the medium, giving the neutral addition product.

Aldehydes react faster than ketones — two reasons

- **Steric effect.** A ketone has **two** alkyl groups next to the carbonyl, an aldehyde has one (or none). The crowding blocks the nucleophile's approach.
- **Electronic effect.** Alkyl groups push electrons (+I), which **reduces the positive charge** on the carbonyl carbon. A ketone has two such groups, so its carbon is less electrophilic.

Both factors favour the aldehyde. The same logic orders any set: the **fewer and smaller the groups on the carbonyl, the faster** the reaction.

Worked example · NCERT intext 8.4 — rank ethanal, propanal, propanone, butanone for nucleophilic addition

Count the groups. Ethanal has one methyl; propanal one ethyl; propanone two methyls; butanone a methyl and an ethyl. Most crowded and most electron-pushing is butanone, then propanone, then propanal, then ethanal. Increasing reactivity: **butanone < propanone < propanal < ethanal**.

Aromatic aldehydes are less reactive than aliphatic ones

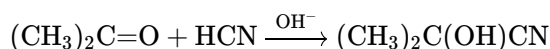
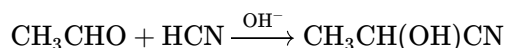
In benzaldehyde the ring pushes electrons into the carbonyl by **resonance** (the $-R$ pull of the carbonyl draws ring electrons towards it), so the carbonyl carbon is **less positive**. Benzaldehyde is therefore less reactive than propanal in nucleophilic addition (2022 Q15(a), 2025 Q1). NCERT example 8.3 and intext 8.4(ii): for aromatic carbonyls the order of reactivity runs from the ketone (acetophenone), through *p*-tolualdehyde (a methyl donates), benzaldehyde, to ***p*-nitrobenzaldehyde** (the nitro group withdraws, so the carbon is most positive and reacts fastest).

Trap: do not answer "benzaldehyde is less reactive because it is bulky". The reason is **resonance with the ring lowering the positive charge** on the carbonyl carbon, and you should say so, with "less electrophilic".

Nucleophilic addition: HCN, bisulphite, Grignard — *three reagents that add across C=O and give three different products*

Hydrogen cyanide → cyanohydrin

[exposure] A **cyanohydrin** carries an $-OH$ and a $-CN$ on the same carbon. Pure HCN is slow, so a trace of **base** is added: it removes H^+ from HCN and gives the cyanide ion CN^- , a much stronger nucleophile. CN^- attacks the carbonyl carbon, and the oxygen then takes H^+ . Cyanohydrins are useful synthetic intermediates.



Sodium hydrogensulphite → addition product

NaHSO₃ adds to give a crystalline **hydrogensulphite addition compound**. For most aldehydes the equilibrium lies to the product side; for most ketones it lies to the reactant side, because of steric hindrance. The compound is water-soluble. Warm it with dilute mineral acid or alkali and the carbonyl compound is regenerated — so the reaction is used to **separate and purify aldehydes**: make the adduct, filter away the insoluble impurities, then decompose the adduct (2023 Q33(b)).

Grignard reagent → alcohol

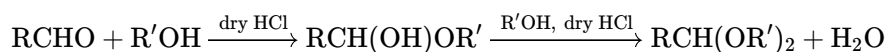
The R^{δ-} of a Grignard reagent attacks the carbonyl carbon; hydrolysis then gives an alcohol. What you get depends on the carbonyl you started with:

Carbonyl	With RMgX, then H ₂ O
Methanal HCHO	primary alcohol RCH ₂ OH
Any other aldehyde	secondary alcohol RCH(OH)R'
A ketone	tertiary alcohol RR'C(OH)R''

Example (2022 Q8): propanone + CH₃MgI, then water, gives 2-methylpropan-2-ol (tertiary). Ethanal + CH₃MgBr gives propan-2-ol (secondary).

Nucleophilic addition: alcohols — *hemiacetals, acetals and ketals* — and why dry HCl is there

[exposure] An **aldehyde** reacts with **one** equivalent of a monohydric alcohol (one –OH) in the presence of **dry HCl** to give a **hemiacetal**: a carbon holding both an –OH and an –OR. With a second equivalent of alcohol the –OH is replaced by another –OR, giving an **acetal** (a **gem-dialkoxy** compound: two –OR groups on the same carbon).



A **ketone** does not react with a monohydric alcohol, but it does react with **ethylene glycol** (HOCH₂CH₂OH, two –OH groups) under the same conditions, forming a cyclic product, an **ethylene glycol ketal**.

Why dry HCl: it **protonates the carbonyl oxygen**, which makes the carbonyl carbon more positive (more electrophilic), so the weakly nucleophilic alcohol can attack. Acetals and ketals are reversed by **dilute aqueous mineral acid**, which gives back the aldehyde or ketone.

Examiner asks: "draw the structure of the hemiacetal of ethanal" and "cyanohydrin of cyclobutanone" (2020 Q23(b)).

Nucleophilic addition–elimination: ammonia and its derivatives — $\text{H}_2\text{N}-\text{Z}$ attacks, water leaves, and $\text{C}=\text{N}$ appears

Ammonia and its derivatives $\text{H}_2\text{N}-\text{Z}$ (Z may be an alkyl or aryl group, $-\text{OH}$, $-\text{NH}_2$, $-\text{NHC}_6\text{H}_5$, $-\text{NHCONH}_2$...) add to $\text{C}=\text{O}$ with acid as a catalyst, and then **lose water** by rapid dehydration of the intermediate. The net change is $\text{C}=\text{O} \longrightarrow \text{C}=\text{N}-\text{Z}$. Because the dehydration is fast, the equilibrium is driven to the product.

Reagent $\text{H}_2\text{N}-\text{Z}$	Product of $\text{C}=\text{O}$
Ammonia NH_3	imine $\text{C}=\text{NH}$
A primary amine RNH_2	substituted imine, Schiff's base $\text{C}=\text{NR}$
Hydroxylamine NH_2OH	oxime $\text{C}=\text{N}-\text{OH}$
Hydrazine NH_2NH_2	hydrazone $\text{C}=\text{N}-\text{NH}_2$
Phenylhydrazine $\text{C}_6\text{H}_5\text{NHNH}_2$	phenylhydrazone
2,4-Dinitrophenylhydrazine (2,4-DNP)	2,4-dinitrophenylhydrazone — a yellow, orange or red solid
Semicarbazide $\text{NH}_2\text{NHCONH}_2$	semicarbazone

Aldehyde products take the suffix by name: from ethanal you get acetaldehyde oxime, acetaldehyde hydrazone, and so on; from a ketone, acetone semicarbazone. The derivatives are **crystalline solids with sharp melting points**, so they are used to **identify** a carbonyl compound — and the **2,4-DNP test** (a yellow/orange/red precipitate) is the standard way to show that a compound is an aldehyde or ketone at all.

Trap: semicarbazide has **two** $-\text{NH}_2$ groups, but only **one** forms the semicarbazone. The lone pair on the nitrogen next to $\text{C}=\text{O}$ is in resonance with it, so that nitrogen is not nucleophilic enough; the terminal NH_2 , whose lone pair is free, does the attacking (2014 Q13(b)).

Examiner asks: NCERT intext 8.5 (predict the products: cyclopentanone + NH_2OH gives the oxime; cyclohexanone + 2,4-DNP gives the 2,4-dinitrophenylhydrazone; a compound with both $\text{C}=\text{C}$ and CHO and semicarbazide still keeps the $\text{C}=\text{C}$); the semicarbazone of ethanal (2022 Q27(c)).

Reduction of aldehydes and ketones — to alcohols, or all the way down to hydrocarbons

To alcohols

NaBH_4 , LiAlH_4 or catalytic hydrogenation (H_2 with Ni, Pt or Pd) reduce an **aldehyde to a primary alcohol** and a **ketone to a secondary alcohol**. $\text{CH}_3\text{CHO} \xrightarrow{\text{H}_2 / \text{Pt}} \text{CH}_3\text{CH}_2\text{OH}$;
 $\text{CH}_3\text{COCH}_3 \xrightarrow{\text{LiAlH}_4} \text{CH}_3\text{CH}(\text{OH})\text{CH}_3$.

To hydrocarbons — remove the oxygen completely

The $\text{C}=\text{O}$ becomes CH_2 in either of two reactions, one in acid and one in base:

- **Clemmensen reduction** — zinc amalgam and concentrated HCl .
 $\text{CH}_3\text{CH}_2\text{CHO} \xrightarrow{\text{Zn-Hg} / \text{conc. HCl}} \text{CH}_3\text{CH}_2\text{CH}_3$. [**exposure**] **Zinc amalgam** is zinc alloyed with mercury; the metal and acid together make the hydrogen that does the reducing.
- **Wolff–Kishner reduction** — hydrazine first (giving the hydrazone, losing water), then heating with KOH or NaOH in a **high-boiling solvent** such as ethylene glycol; nitrogen gas escapes and hydrogen goes on in its place. $\text{CH}_3\text{COCH}_3 \xrightarrow{\text{NH}_2\text{NH}_2} (\text{CH}_3)_2\text{C}=\text{NNH}_2 \xrightarrow{\text{KOH, ethylene glycol, } \Delta} \text{CH}_3\text{CH}_2\text{CH}_3 + \text{N}_2$.

Trap: choose by what else is in the molecule. Clemmensen uses strong acid, Wolff–Kishner strong base — pick the one that the rest of the molecule survives. For a plain "reduce the carbonyl to CH₂" question, either is accepted; name the one you write.

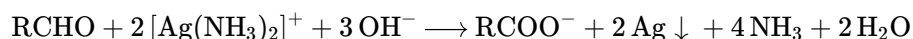
Examiner asks: Wolff–Kishner and Clemmensen as "explain/write the reaction" questions (2022 Q11, Q28(b), 2025 Q6(b)).

Oxidation: the tests that tell aldehydes from ketones — *Tollens', Fehling's, and the haloform reaction*

Aldehydes oxidise easily to carboxylic acids with the *same* number of carbons, even with **mild** oxidants. They can do so because the carbonyl carbon carries a **hydrogen** that can turn into –OH without any bond being broken. **Ketones** have no such hydrogen. They oxidise only under vigorous conditions (a strong oxidant and heat), by breaking a carbon–carbon bond next to the carbonyl; the result is a **mixture of acids with fewer carbons**. This difference is the basis of the distinguishing tests.

Tollens' test

[exposure] Tollens' reagent is **ammoniacal silver nitrate**, [Ag(NH₃)₂]⁺ in alkaline solution, always made fresh. It is a mild oxidant. Warmed with an **aldehyde**, it is reduced to silver metal, which coats the tube as a **bright silver mirror**; the aldehyde becomes a carboxylate. A ketone gives nothing.



Fehling's test

[exposure] Fehling's reagent is two solutions mixed in equal amounts just before use: **Solution A**, aqueous copper(II) sulphate, and **Solution B**, alkaline sodium potassium tartrate (**Rochelle salt**). Warmed with an aldehyde, it gives a **reddish-brown precipitate** of copper(I) oxide Cu₂O. Ketones do not react, and **aromatic aldehydes (benzaldehyde) do not respond** to Fehling's, because it is too weak an oxidant for them.

The haloform (iodoform) reaction

Any aldehyde or ketone with a **methyl group on the carbonyl carbon**, CH₃–CO– (a **methyl ketone**), is oxidised by sodium hypohalite NaOX. The methyl group leaves as a **haloform** CHX₃ and the rest becomes a carboxylate with **one carbon fewer**.



For the **iodoform test** you use I₂ with NaOH (which makes NaOI in the flask; "NaOH + I₂" in a question means the same). A **yellow precipitate** of CHI₃ (iodoform) shows a CH₃CO– group, or a CH₃CH(OH)– group, which is first oxidised to CH₃CO–. The reaction does not touch a C=C.

Worked example · NCERT Example 8.4 — compound A, C₈H₈O

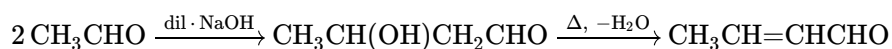
A gives an orange-red precipitate with 2,4-DNP, so it is a carbonyl compound. It gives a yellow precipitate with NaOH / I₂, so it carries CH₃CO–. It does **not** reduce Tollens' or Fehling's, so it is a **ketone**. It does not decolourise bromine water or Baeyer's reagent, so the unsaturation in the formula is not an alkene; it is an aromatic ring. The formula fits a ring bearing COCH₃: **acetophenone**. Drastic oxidation gives benzoic acid C₇H₆O₂, and the iodoform reaction gives sodium benzoate and CHI₃.

Examiner asks: distinguish an aldehyde from a ketone (2020 Q23(a), 2017 Q16(b), 2025 Q7(b)); the same acetophenone puzzle (2025 Q3); the iodoform reaction in conversions (2017 Q18(a), 2023 Q32).

α -Hydrogen reactions: aldol condensation — why the hydrogens next to C=O are acidic, and what that lets two molecules do

[exposure] The carbon next to the carbonyl is the **α -carbon**, and the hydrogens on it are **α -hydrogens**. They are noticeably **acidic** — a base can remove one. Two reasons: the carbonyl is strongly **electron-withdrawing**, which weakens the C–H bond; and the resulting carbanion is **resonance-stabilised**, because its negative charge can sit on the carbonyl oxygen. (2023 Q34(b))

[exposure] Aldol condensation. An aldehyde or ketone with at least one α -hydrogen, in **dilute alkali**, joins two of its molecules. The base takes an α -hydrogen to make a carbanion; that carbanion attacks the carbonyl carbon of a second molecule; the oxygen then takes H^+ . The product of an **aldehyde** is a **β -hydroxy aldehyde**, an **aldol** (aldehyde + alcohol); that of a **ketone** is a **β -hydroxy ketone**, a **ketol**. On **heating** they lose water to give an **α,β -unsaturated carbonyl compound**, the aldol condensation product.



That is ethanal $\rightarrow$ 3-hydroxybutanal (the aldol) $\rightarrow$ but-2-enal. Propanone gives 4-hydroxy-4-methylpentan-2-one (the ketol) and then 4-methylpent-3-en-2-one on heating.

Cross aldol condensation

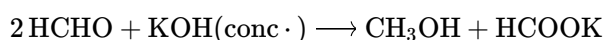
Two *different* carbonyl compounds, both with α -hydrogens, give a mixture of **four** products — two "self" and two "cross". Ethanal + propanal gives: but-2-enal and 2-methylpent-2-enal (self), and 2-methylbut-2-enal and pent-2-enal (cross). If one partner has **no α -hydrogen** (such as benzaldehyde), it can only be the acceptor and a single product forms: benzaldehyde + acetophenone gives **1,3-diphenylprop-2-en-1-one** (benzalacetophenone).

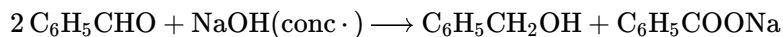
Trap: the condition for aldol is **at least one α -hydrogen**. Benzaldehyde, methanal and $(\text{CH}_3)_3\text{C}-\text{CHO}$ have none and cannot self-aldol; they do the Cannizzaro reaction instead (next section). Do not forget to heat if the question asks for the condensation product (the α,β -unsaturated compound), not just the aldol.

Examiner asks: describe aldol condensation (2015 Q17(b)); ethanal to but-2-enal (2022 Q24(c)); $\text{CH}_3\text{CN} \rightarrow$ ethanal $\rightarrow$ but-2-enal (2017 Q26); the 2025 puzzle where B is made from ethanal and dilute NaOH (2025 Q2); cross-aldol of benzaldehyde with ethanal (2025 Q4(b)).

Cannizzaro reaction; ring substitution; uses — what an aldehyde with no α -hydrogen does, and where the carbonyl directs a ring

[exposure] Cannizzaro reaction. An aldehyde with **no α -hydrogen**, heated with **concentrated alkali**, **reacts with itself**: one molecule is **oxidised** to the carboxylate salt, the other is **reduced** to the alcohol. A reaction in which one substance is both oxidised and reduced is a **disproportionation**.





Electrophilic substitution in the ring. In an aromatic aldehyde or ketone the carbonyl pulls electrons out of the ring (the $-R$ effect) and puts a partial positive charge at the ortho and para positions, so it is **deactivating** and **meta-directing**. Benzaldehyde + $\text{HNO}_3 / \text{H}_2\text{SO}_4$ at 273–283 K gives **3-nitrobenzaldehyde** (*m*-nitrobenzaldehyde).

Uses. 40% aqueous methanal is **formalin**, a preservative for biological specimens; methanal is also the monomer for **Bakelite** (phenol–formaldehyde resin) and urea–formaldehyde glues. Ethanal is the starting material for ethanoic acid, ethyl ethanoate, vinyl acetate, polymers and drugs. Benzaldehyde goes into perfumes and dyes; propanone and butanone are industrial solvents; butyraldehyde, vanillin, acetophenone and camphor are used for odour and flavour.

Examiner asks: the Cannizzaro reaction (2022 Q11, 2013 Q19(a), 2025 Q5(a)); the compound that gives sodium formate and methanol (Q29 in the 30+ video — methanal); nitration of benzaldehyde (2013 Q19(b)).

Carboxylic acids: structure and names — *the carboxyl group* — COOH , and a table of names you must know cold

[exposure] A **carboxylic acid** carries the **carboxyl group** — COOH : a carbonyl and a hydroxyl on the same carbon (the name is "carbonyl" + "hydroxyl"). It is the one family in this chapter that is an **acid**, which is why it turns blue litmus red. If the group sits on an alkyl chain the acid is **aliphatic**; if it sits on a benzene ring it is **aromatic**. The higher aliphatic acids (C_{12} – C_{18}) are **fatty acids**; palmitic acid $\text{C}_{15}\text{H}_{31}\text{COOH}$ is an example. In natural fats they are present as esters of glycerol.

Common names — where they come from

Carboxylic acids were among the first organic compounds isolated, so most carry a common name from the Latin or Greek for their source: **formic** acid (from *formica*, ant), **acetic** acid (*acetum*, vinegar), **butyric** acid (*butyrum*, butter — it is in rancid butter).

IUPAC names

- One** — COOH : replace the final "e" of the alkane with **-oic acid** (the carboxyl carbon is C_1): methanoic, ethanoic, propanoic acid.
- Two** — COOH **groups**: both carbons are counted in the chain, and the "e" is **kept**: propane-1,3-dioic acid = **propanedioic acid** (no locants needed when they sit at both ends).
- Three or more** — COOH **groups**: the carboxyl carbons are **not** counted; they are named as **carboxylic acids** on the backbone: propane-1,2,3-tricarboxylic acid.
- On a ring**: ring name + **carboxylic acid**: 2-methylcyclopentanecarboxylic acid. $\text{C}_6\text{H}_5\text{COOH}$ is benzoic acid, also benzenecarboxylic acid.

Structure	Common name	IUPAC name
HCOOH	formic acid	methanoic acid
CH ₃ COOH	acetic acid	ethanoic acid
CH ₃ CH ₂ COOH	propionic acid	propanoic acid
CH ₃ CH ₂ CH ₂ COOH	butyric acid	butanoic acid
(CH ₃) ₂ CHCOOH	isobutyric acid	2-methylpropanoic acid
HOOC–COOH	oxalic acid	ethanedioic acid
HOOC–CH ₂ –COOH	malonic acid	propanedioic acid
HOOC–(CH ₂) ₂ –COOH	succinic acid	butanedioic acid
HOOC–(CH ₂) ₃ –COOH	glutaric acid	pentanedioic acid
HOOC–(CH ₂) ₄ –COOH	adipic acid	hexanedioic acid
C ₆ H ₅ COOH	benzoic acid	benzenecarboxylic acid
C ₆ H ₅ CH ₂ COOH	phenylacetic acid	2-phenylethanoic acid
benzene with two adjacent –COOH	phthalic acid	benzene-1,2-dicarboxylic acid

More examples (NCERT intext 8.6): C₆H₅CH₂CH₂COOH = 3-phenylpropanoic acid; (CH₃)₂C=CHCOOH = 3-methylbut-2-enoic acid; 2,4,6-trinitrobenzoic acid.

Structure of the carboxyl group — *why the carbonyl carbon here is less electrophilic than in an aldehyde*

The carboxyl carbon is *sp*² hybridised and **trigonal planar** (all bonds in one plane, about 120°). But it is **less electrophilic** than the carbonyl carbon of an aldehyde or ketone. The reason is **resonance**: a lone pair on the –OH oxygen flows towards the carbon, so the carbon receives electron density from the oxygen and its partial positive charge is smaller. A nucleophile therefore has much less to attack.

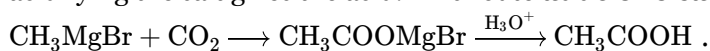
Trap: this is the standard answer to "why do carboxylic acids not give the characteristic reactions of the carbonyl group" (2017 Q6, 2025 Q5(b)(i)). Write: the lone pair on the –OH oxygen is in **resonance** with the C=O, which **reduces the electrophilicity** of the carbon, so the nucleophilic-addition reactions do not occur.

Preparing carboxylic acids — *six routes; the carbon count tells you which one*

- Oxidising primary alcohols and aldehydes.** Use a **strong** oxidant: KMnO₄ (acidic, alkaline or neutral), K₂Cr₂O₇ (acidic), or CrO₃ in acid, known as the **Jones reagent**. (Anhydrous CrO₃ without the acid is mild.) Aldehydes also oxidise with the *mild* Tollens' and Fehling's reagents.
- Oxidising an alkylbenzene.** Vigorous oxidation with hot alkaline KMnO₄ (or chromic acid) turns the **whole side chain, however long, into –COOH**: toluene, ethylbenzene and propylbenzene all give potassium benzoate, and acidification gives **benzoic acid** (2019 Q4).
- Hydrolysing nitriles and amides.** RCN with H⁺ or OH[–] goes to the amide and then to the acid. Mild conditions stop at the amide; complete hydrolysis gives the acid. An amide heated with water gives

the acid plus NH_3 .

4. **Grignard reagent and dry ice.** RMgX attacks the carbon of solid CO_2 (dry ice) in dry ether; acidifying the salt gives the acid. This route **adds one carbon**:

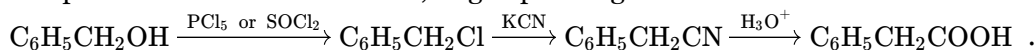


5. **Hydrolysing acid chlorides and anhydrides.** Water gives the acid; aqueous base is faster and gives the carboxylate, which you acidify.

6. **Hydrolysing esters.** Acid hydrolysis gives acid + alcohol directly. Base hydrolysis is faster and gives the carboxylate salt, which you acidify. $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOC}_2\text{H}_5 \xrightarrow{\text{NaOH}} \text{CH}_3\text{CH}_2\text{CH}_2\text{COONa} + \text{C}_2\text{H}_5\text{OH}.$

Worked example · NCERT Example 8.5 — benzyl alcohol to phenylethanoic acid

The product has one carbon more, so go up through the nitrile.



Worked example · NCERT Example 8.5 — 3-nitrobromobenzene to 3-nitrobenzoic acid

Br must become COOH , one carbon added: make the Grignard reagent (Mg, dry ether), add CO_2 , hydrolyse. The NO_2 survives untouched.

Two more NCERT conversions: butan-1-ol $\rightarrow$ butanoic acid with acidic $\text{K}_2\text{Cr}_2\text{O}_7$; 4-methylacetophenone $\rightarrow$ benzene-1,4-dicarboxylic acid with KMnO_4 / KOH (both the methyl and the ketone are oxidised).

Cyclohexene $\rightarrow$ hexanedioic acid with hot KMnO_4 / H_2SO_4 breaks the ring at the double bond.

Examiner asks: ethylbenzene + KMnO_4/KOH then acid (2019 Q4; 2025 Q4(a)); benzene-ring conversions to benzoic acid (NCERT intext 8.7); the Grignard- CO_2 route (2017 Q26).

Physical properties of carboxylic acids — *hydrogen-bonded dimers explain the high boiling points*

- **State and smell.** The first three aliphatic acids are colourless liquids with a sharp smell; the higher ones are waxy solids with almost no smell (low volatility).
- **Boiling point.** Higher than aldehydes, ketones and even alcohols of similar mass. **Reason:** carboxylic acid molecules hydrogen-bond strongly in pairs. The $\text{O}-\text{H}$ of an acid is **more polar** than that of an alcohol, because the carbonyl group next to it pulls electrons away, so the hydrogen bonds are stronger.
- **[exposure] A dimer** is a pair of identical molecules held together; here, two acid molecules joined by two hydrogen bonds. The bonds are so strong that most carboxylic acids stay as dimers **even in the vapour phase** and in **aprotic solvents** (solvents that have no acidic hydrogen to give away).
- **Solubility.** Aliphatic acids up to **four carbons** mix with water (hydrogen bonds to water). Solubility drops as the carbon chain grows. Benzoic acid, the simplest aromatic acid, is insoluble in cold water because of the large hydrophobic ring. Acids dissolve in less polar organic solvents such as benzene, ether, alcohol and chloroform.

Examiner asks: which of ethanoic acid and ethanol has the higher boiling point, and why (2023 Q35(b): ethanoic acid, stronger hydrogen bonding and dimer formation).

Acidity: what carboxylic acids do with metals, alkali and carbonates — *the O–H bond breaks; the test with bicarbonate is the one examiners love*

Carboxylic acids are fairly strong acids because the $\text{O}-\text{H}$ bond is polar and the **carboxylate ion** RCOO^- left behind is stabilised by resonance (below). They turn blue litmus red and react with:

- **Active metals** (Na, K, Mg): $2 \text{RCOOH} + 2 \text{Na} \longrightarrow 2 \text{RCOONa} + \text{H}_2$. (Alcohols do this too.)
- **Alkalis:** $\text{RCOOH} + \text{NaOH} \longrightarrow \text{RCOONa} + \text{H}_2\text{O}$ (neutralisation). Phenols do this too.
- **Carbonates and bicarbonates:** $\text{RCOOH} + \text{NaHCO}_3 \longrightarrow \text{RCOONa} + \text{H}_2\text{O} + \text{CO}_2 \uparrow$ with a **brisk effervescence** of CO_2 . Alcohols and phenols are too weak to do it.

Trap: the **sodium bicarbonate test** is the standard test to detect a carboxyl group, and the answer to every "distinguish ethanoic acid from ethanol", "benzoic acid from phenol", "benzoic acid from ethyl benzoate" question (2017 Q12(b), 2023 Q34(a), 2025 Q7(b)(i)). Write the observation: brisk effervescence of CO_2 with the acid, nothing with the other.

Putting a number on strength: K_a and pK_a

[exposure] For $\text{RCOOH} + \text{H}_2\text{O} \rightleftharpoons \text{RCOO}^- + \text{H}_3\text{O}^+$ the **acid dissociation constant** is $K_a = \frac{[\text{RCOO}^-][\text{H}_3\text{O}^+]}{[\text{RCOOH}]}$ (the constant concentration of water is folded in), and $pK_a = -\log K_a$. A **larger** K_a means a stronger acid, but the numbers are awkward powers of ten, so the pK_a is used. **The smaller the pK_a , the stronger the acid.** NCERT's bands: below 1, a **strong** acid; 1 to 5, **moderately strong**; 5 to 15, **weak**; above 15, **extremely weak**.

Why carboxylic acids beat alcohols and phenols — *it is all about how stable the left-over negative ion is*

A stronger acid is one whose **conjugate base** (the ion left after H^+ leaves) is more stable, because then the equilibrium lies further towards the ions. Compare the three:

- **Carboxylic acid vs alcohol.** The carboxylate RCOO^- is stabilised by **resonance**: its two structures are **identical**, and the negative charge is spread over **two oxygens**. The alkoxide RO^- of an alcohol has no resonance, and the electron-pushing R group makes its single oxygen even more crowded with charge. So carboxylic acids are much stronger than alcohols. (Ethanoic acid versus ethanol is asked as 2025 Q5(b)(ii).)
- **Carboxylic acid vs phenol.** The phenoxide ion also has resonance, but its charge is shared between **one** oxygen and less electronegative ring carbons. The carboxylate puts the charge on **two electronegative oxygens**, so delocalisation is more effective. So carboxylic acids are stronger than phenols.

The acid itself is also stabilised by resonance, but less so: one of its contributing structures has **charge separation** (a positive and a negative atom), which costs energy.

Ranking: **carboxylic acid > phenol > alcohol**. All are weaker than mineral acids.

What a substituent does to acid strength — *electron-pullers raise it, electron-pushers lower it — four rules*

The same logic: anything that **stabilises the carboxylate** makes a stronger acid.

- **Electron-withdrawing group (EWG)** — $-\text{NO}_2$, $-\text{CN}$, halogens, $-\text{CF}_3$ — pulls charge towards itself and **stabilises the carboxylate**: stronger acid. An **electron-releasing group (ERG)** — alkyl, $-\text{OCH}_3$ — pushes charge onto the carboxylate and **destabilises** it: weaker acid.
- **Number of substituents:** more is stronger. $\text{CH}_3\text{COOH} < \text{ClCH}_2\text{COOH} < \text{Cl}_2\text{CHCOOH} < \text{Cl}_3\text{CCOOH}$

- **Nature of the substituent:** the more electronegative, the stronger. $\text{CH}_3\text{COOH} < \text{ICH}_2\text{COOH} < \text{BrCH}_2\text{COOH} < \text{ClCH}_2\text{COOH} < \text{FCH}_2\text{COOH}$. Fluorine is the strongest puller.
- **Position:** the **closer** the EWG is to the $-\text{COOH}$, the stronger the effect, because its pull falls off with distance. 2-fluorobutanoic acid > 3-fluorobutanoic acid > 4-fluorobutanoic acid.

A surprising one: a **phenyl or vinyl group** attached directly to the carboxyl carbon **increases** acidity relative to a saturated acid. You might expect resonance donation to weaken it, but the carbon the carboxyl group is attached to is sp^2 , which is more electronegative than an sp^3 carbon, and that wins.

Worked example · the benzoic acid series (pK_a from NCERT)

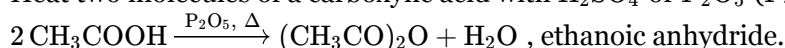
4-Methoxybenzoic acid (pK_a 4.46) < benzoic acid (4.19) < 4-nitrobenzoic acid (3.41) in acid strength. $-\text{OCH}_3$ donates electrons and weakens the acid; $-\text{NO}_2$ withdraws and strengthens it. A lower pK_a means a stronger acid. 3,4-Dinitrobenzoic acid, with two NO_2 groups, is the strongest of all four (2022 Q21(b)).

Examiner asks: "why is ethanoic acid a stronger acid than ethanol" (2025 Q5(b)(ii)); "4-nitrobenzoic acid is a stronger acid than benzoic acid" (2022 Q15(b)); "arrange CH_3COOH , ClCH_2COOH , FCH_2COOH by acid strength" (2022 Q5(a)); NCERT intext 8.8, which asks which of each pair is stronger.

Carboxylic acid reactions: breaking the C–OH bond — *anhydride, ester, acid chloride, amide* — the four derivatives made from the acid

Anhydride

Heat two molecules of a carboxylic acid with H_2SO_4 or P_2O_5 (P_4O_{10}) and water is removed:



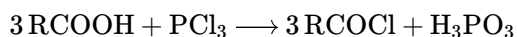
Ester — esterification

[exposure] Esterification is making an ester from a carboxylic acid and an alcohol (or phenol), with an acid catalyst (conc. H_2SO_4 or HCl gas): $\text{RCOOH} + \text{R}'\text{OH} \xrightleftharpoons{\text{H}^+} \text{RCOOR}' + \text{H}_2\text{O}$. It is reversible; esters smell fruity.

Mechanism (all you need is the sequence): (1) H^+ from the catalyst **protonates the carbonyl oxygen**, which makes the carbon more positive. (2) The alcohol's oxygen, with its lone pair, **attacks that carbon** (nucleophilic attack), giving a tetrahedral intermediate. (3) A **proton transfers** between oxygens, so one $-\text{OH}$ becomes $-\text{OH}_2^+$. (4) **Water leaves**, giving the protonated ester. (5) It **loses** H^+ , giving the ester and handing back the catalyst. The H^+ is used and regained, so it is a catalyst.

Acid chloride

The $-\text{OH}$ is replaced by $-\text{Cl}$ with PCl_5 , PCl_3 or SOCl_2 (thionyl chloride):



Thionyl chloride is preferred: its by-products, SO_2 and HCl , are gases that leave the mixture, so the acid chloride comes out pure.

Amide

A carboxylic acid with NH_3 gives the ammonium salt; **heating** the salt loses water to give the amide.

$\text{CH}_3\text{COOH} + \text{NH}_3 \longrightarrow \text{CH}_3\text{COO}^-\text{NH}_4^+ \xrightarrow{\Delta} \text{CH}_3\text{CONH}_2 + \text{H}_2\text{O}$. Phthalic acid takes two NH_3 to give ammonium phthalate, which on heating gives **phthalamide**; strong heating drives off a molecule of NH_3 and closes the ring to **phthalimide**.

Examiner asks: CH_3COOH with P_4O_{10} (2025 Q6(a)); phthalamide on strong heating gives phthalimide (2025 Q6(a)); acid + NH_3 then heat gives the amide, which is the middle step of the Hofmann chains you will meet in Amines.

Carboxylic acid reactions: reduction, decarboxylation, hydrocarbon part — LiAlH_4 reaches the acid; soda lime removes CO_2 ; Br₂/red P halogenates the α -carbon

Reduction

LiAlH_4 in ether, or better **diborane** B_2H_6 , reduces a carboxylic acid to a **primary alcohol**. Diborane does not easily reduce esters, nitro or halo groups, so it is selective. **Sodium borohydride** NaBH_4 is too mild and does not reduce a carboxyl group.

Decarboxylation

[exposure] Decarboxylation removes CO_2 from a carboxylic acid, leaving a hydrocarbon with **one carbon less**. The sodium salt of the acid is heated with **soda lime** ($\text{NaOH} + \text{CaO}$ in a **3 : 1** ratio):

$\text{RCOONa} + \text{NaOH} \xrightarrow{\text{CaO}, \Delta} \text{RH} + \text{Na}_2\text{CO}_3$. Examples: sodium ethanoate gives **methane**; sodium benzoate gives **benzene**.

[exposure] Kolbe electrolysis. Electrolysing the **aqueous sodium or potassium salt** of an acid is also a decarboxylation, but the alkyl groups from **two** carboxylate ions join: at the anode

$2\text{RCOO}^- \longrightarrow \text{R-R} + 2\text{CO}_2 + 2\text{e}^-$ and at the cathode $2\text{H}_2\text{O} + 2\text{e}^- \longrightarrow \text{H}_2 + 2\text{OH}^-$. The hydrocarbon has **twice** the carbons of the alkyl group of the acid.

Hell–Volhard–Zelinsky (HVZ) reaction

A carboxylic acid with Cl_2 or Br_2 and a little **red phosphorus**, then water, gets a halogen on the **α -carbon**, giving an **α -halocarboxylic acid**. $\text{CH}_3\text{CH}_2\text{COOH} \xrightarrow[\text{H}_2\text{O}]{\text{Br}_2 / \text{red P}} \text{CH}_3\text{CHBrCOOH}$ (2-bromopropanoic acid).

Aqueous NaOH then replaces the halogen by $-\text{OH}$ (ethanoic acid $\rightarrow$ 2-chloroethanoic acid $\rightarrow$ 2-hydroxyethanoic acid, 2017 Q18(b)).

The benzene ring of an aromatic acid

The carboxyl group is **deactivating and meta-directing** (the $-\text{R}$ effect pulls ring electrons, leaving the ortho/para positions positive). Benzoic acid + $\text{HNO}_3 / \text{H}_2\text{SO}_4$ gives **3-nitrobenzoic acid**; with $\text{Br}_2 / \text{FeBr}_3$ it gives 3-bromobenzoic acid. It does **not undergo Friedel–Crafts**, for two reasons: the ring is deactivated, and the AlCl_3 catalyst bonds to the electron-rich carboxyl oxygens instead of working as a catalyst (2022 Q14(a), 2023 Q33(a)).

Uses

Methanoic acid goes into rubber, textile, dyeing, leather and electroplating; ethanoic acid is a solvent and the acid of vinegar; hexanedioic acid is a monomer for **nylon-6,6**; esters of benzoic acid are used in perfumery; sodium benzoate is a food preservative; higher fatty acids go into soaps and detergents.

Examiner asks: HVZ (2019 Q7, 2022 Q24(b)); decarboxylation with soda lime (2017 Q12(a), 2022 Q28(c), 2023 Q35, 2025 Q6(b)); the conversion that uses LiAlH_4 vs NaBH_4 .

Patterns, collected — *the reagents, tests and orderings you will actually be asked to produce*

1. Which reagent for which job

To make	From	Reagent
aldehyde	primary alcohol	PCC (or Cu, 573 K)
ketone	secondary alcohol	anhydrous CrO_3 (or Cu, 573 K)
aldehyde	acid chloride	H_2 / Pd– BaSO_4 (Rosenmund)
aldehyde	nitrile	SnCl_2 / HCl then H_3O^+ (Stephen), or DIBAL-H
benzaldehyde	toluene	CrO_2Cl_2 (Etard); CrO_3 / $(\text{CH}_3\text{CO})_2\text{O}$; or Cl_2 , $h\nu$ then hydrolysis
ketone	acid chloride	R_2Cd
ketone	benzene	RCOCl / AlCl_3 (Friedel–Crafts)
carboxylic acid (+1 C)	alkyl halide	Mg, CO_2 , H_3O^+ — or KCN then hydrolysis
acid (same C)	primary alcohol / aldehyde	KMnO_4 or $\text{K}_2\text{Cr}_2\text{O}_7$ / H^+
hydrocarbon (one C less)	acid salt	soda lime, heat

2. Distinguish by a test

Test	Positive for	Negative for
Tollens' (silver mirror)	aldehyde	ketone
Fehling's (red-brown Cu_2O)	aliphatic aldehyde	ketone, <i>aromatic</i> aldehyde
Iodoform ($\text{NaOH} + \text{I}_2$, yellow CHI_3)	$\text{CH}_3\text{CO}-$ or $\text{CH}_3\text{CH}(\text{OH})-$	others
2,4-DNP (orange ppt)	any aldehyde or ketone	acids, esters, alcohols
NaHCO_3 (effervescence)	carboxylic acid	alcohol, phenol, ester

Worked example

Distinguish propanal from propanone (2020 Q23(a)): warm each with freshly prepared Tollens' reagent. Propanal gives a bright silver mirror ($\text{CH}_3\text{CH}_2\text{CHO} \longrightarrow \text{CH}_3\text{CH}_2\text{COO}^-$, $\text{Ag}^+ \longrightarrow \text{Ag}$); propanone does not react.

- Distinguish ethanol from ethanoic acid — answer only: add NaHCO_3 ; the acid gives brisk effervescence of CO_2 , the alcohol does not
- Distinguish benzaldehyde from acetophenone — answer only: Tollens' gives a silver mirror with benzaldehyde; iodoform test (yellow ppt) is given only by acetophenone

3. Carbon count: how it changes

Worked example

Which reactions change the carbon count? **Gain one:** Grignard + CO_2 ; KCN then hydrolysis; nitrile route.

Lose one: haloform (methyl ketone $\rightarrow$ acid, CHX_3 lost); soda-lime decarboxylation; Hofmann (Chapter 9).

Double: aldol condensation ($2 \text{CH}_3\text{CHO} \rightarrow \text{C}_4$); Kolbe electrolysis.

- Ethanal to but-2-enal (carbons: $2 \rightarrow 4$) — answer only: aldol condensation, dilute NaOH then heat
- Propanoic acid to 2-bromopropanoic acid — answer only: Br_2 / red P then water (HVZ)

4. Rank by reactivity, acidity or boiling point

Worked example

Increasing acid strength: 4-methoxybenzoic acid, benzoic acid, 4-nitrobenzoic acid, 3,4-dinitrobenzoic acid (2022 Q21(b)). More electron-withdrawing groups give a stronger acid, and an electron donor ($-\text{OCH}_3$) a weaker one: **4-methoxybenzoic < benzoic < 4-nitrobenzoic < 3,4-dinitrobenzoic.**

- Increasing nucleophilic-addition reactivity: ethanal, propanone, propanal, butanone — answer only: butanone < propanone < propanal < ethanal
- Increasing boiling point: propane, ethanal, ethanol — answer only: propane < ethanal < ethanol

5. Solve an "Identify A, B, C" puzzle

Worked example · 2023 Q35

A ($\text{C}_4\text{H}_8\text{O}_2$) on acid hydrolysis gives B and C. Oxidising C gives B. The sodium salt of B with soda lime gives methane.

- Soda lime on a sodium salt removes CO_2 : the salt must be CH_3COONa , so B is **ethanoic acid**.
 - Oxidising C gives B, and C is a hydrolysis product with two carbons, so C is **ethanol**.
 - Hydrolysis giving an acid and an alcohol means A is an ester: **ethyl ethanoate** ($\text{CH}_3\text{COOC}_2\text{H}_5$ is $\text{C}_4\text{H}_8\text{O}_2$ ✓).
- $\text{C}_3\text{H}_8\text{O}$ with Cu at 573 K gives B (no Fehling's) which gives a yellow precipitate with I_2/NaOH — answer only: A propan-2-ol; B propanone; the yellow precipitate is CHI_3

Past year questions · question types — ranked by how often they turn up

Two videos feed this section. [30+] is the "30+ PYQ" video for CBSE 2024-25 — Q1–Q35, questions from 2013 to 2023. [2025] is the short "2025 PYQ" video — Q1–Q7, from the 2025 board paper. Question numbers are each video's own numbering, and years come from the video. Where the video gave no year, the question is cited by number only. The CBSE 2026-27 "25 PYQ" video for this chapter was not uploaded when these notes were built; once it is, this section can be refreshed from it. Counts are by question part, from the two videos' lists.

1 · Complete the reaction / write the product / convert — 1 to 5 marks · about 35 parts

Recognise it: "write the product", "complete the reaction", "how will you convert A to B", "give the missing reagent".

- Name the functional group that changes and the bond that forms or breaks.
- Count carbons** in the start and the product: up one $\rightarrow$ KCN or Grignard + CO_2 ; down one $\rightarrow$ haloform or soda lime; doubled $\rightarrow$ aldol or Kolbe.

3. Write each step with its reagent above the arrow; write the structure of every intermediate.

Trap: the reagent is the mark. Rosenmund needs Pd–BaSO₄ (without the BaSO₄ the aldehyde is reduced on to the alcohol); Wolff–Kishner is hydrazine then KOH/ethylene glycol; and a "benzoic acid from alkylbenzene" question always wants KMnO₄/KOH, heat, then acid, whatever the chain length.

2 · Give a reason / account for — 1 to 5 marks · about 12 parts

Recognise it: "account for ...", "why is X more reactive / a stronger acid / unable to ...".

1. State the structural fact (resonance, +I, –I, steric crowding, α-hydrogen, a lone pair).
2. Say how it causes the behaviour in one sentence.

Trap: one answer covers a lot: **the stability of the conjugate base** (carboxylate) explains acid strength; **steric hindrance plus +I** explains aldehyde vs ketone; **resonance lowering the positive charge** explains benzaldehyde vs propanal and why carboxylic acids do not do carbonyl reactions.

3 · Identify the compounds (A, B, C) — 3 to 5 marks · about 9 parts

Recognise it: a formula plus a list of observations (a reagent that gives or does not give a precipitate), and "identify A, B and C".

1. Turn each clue into a group: positive 2,4-DNP → carbonyl; Tollens'/Fehling's positive → aldehyde, negative → ketone; iodoform positive → CH₃CO–; ozonolysis → cut the double bond and join the two carbonyls back.
2. Check the formula after every guess.
3. Write the equation that links the compounds.

Trap: iodoform needs **a methyl on the carbonyl carbon**, not just any methyl. Ethanal is the *only* aldehyde that gives the iodoform test.

4 · Distinguish by a chemical test — 2 or 3 marks · about 6 parts

Recognise it: "give a simple chemical test to distinguish between ...".

1. Pick the test that separates the pair (the table in the Patterns section): aldehyde vs ketone → Tollens' or Fehling's; acid vs the rest → NaHCO₃; methyl ketone → iodoform.
2. State the observation for *both* compounds, including "no reaction".

Trap: an observation is what you *see* (silver mirror, brisk effervescence, yellow precipitate), not the equation. Aromatic aldehydes do not respond to Fehling's.

5 · Describe or write a named reaction — 2 to 5 marks · about 8 parts

Recognise it: "describe acetylation / aldol condensation", "write the equation for Stephen / Etard / Cannizzaro / Wolff–Kishner / decarboxylation".

1. One line of what it is, with the reagents.
2. Then a worked equation with an example, and the product named.

Trap: do not mix the three toluene → benzaldehyde methods. CrO₂Cl₂ in CS₂ is Etard; CrO₃ in acetic anhydride gives benzylidene diacetate; Cl₂ in light gives benzal chloride.

6 · Arrange in order; name or draw — 1 to 2 marks · about 9 parts

Recognise it: "arrange in increasing order of acid strength / boiling point / reactivity in nucleophilic addition", "write the IUPAC name", "draw the structure of ...".

- Orderings: acid strength → more EWG, nearer, stronger; boiling point → hydrogen bonding > dipole–dipole > van der Waals; nucleophilic addition → fewer, smaller groups on the carbonyl, faster.
- Names: the highest-ranking group is the suffix (acid > aldehyde > ketone > alcohol); the others become prefixes (oxo-, hydroxy-).

Trap: read "increasing" or "decreasing" before you write the order; and in the benzoic-acid series the electron donor (4-methoxy) is the *weakest* acid.

Past year questions · mark slots — *what each type is worth*

The 30+ video states its own marks: Q1–Q10 are 1-mark, Q11–Q22 are 2-mark, Q23 onward are 3-mark, except that it names Q33 and Q34 (both 2023) as 2-mark. The 2025 video states Q1 as 2 marks and Q4, Q5, Q6 and Q7 as 5 marks each; for its Q2 and Q3 no marks are stated (inferred: 3 each). So this chapter has appeared as a 5-mark long answer in the 2025 paper — the one place in this set where a 5-marker was recorded.

Question type	1 mark	2 marks	3 marks	5 marks
Complete / product / convert	Q2, 2020 Q3, 2019 Q4, 2019 Q7, Q8, 2019 Q9, Q10 [30+]	2017 Q12(a), 2017 Q18, 2013 Q19, 2015 Q22 [30+]	2022 Q24, 2022 Q28, 2020 Q30, 2017 Q26 [30+]	2025 Q4, Q5, Q6, Q7(a) [2025]
Give a reason	2017 Q6 [30+]	2014 Q13, 2022 Q14, 2022 Q15, 2023 Q33, 2023 Q34(b) [30+]; 2025 Q1 [2025]	—	2025 Q5(b) [2025]
Identify A, B, C	—	—	Q25, Q29, 2023 Q32, 2023 Q35 [30+]; 2025 Q2, Q3 [2025, inferred 3]	2025 Q5(a) [2025]
Distinguish by a test	—	2017 Q12(b), 2017 Q16(b), 2023 Q34(a) [30+]	2020 Q23(a) [30+]	2025 Q7(b) [2025]
Named reaction	2019 Q9 [30+]	2022 Q11, 2015 Q17 [30+]	2015 Q22 [30+]	2025 Q6(b) [2025]
Arrange / name / draw	2014 Q1 [30+]	2015 Q20, 2022 Q5, 2022 Q21 [30+]	—	—

Past year questions · repeat offenders — *appeared more than once — highest probability in the chapter*

2015 Q17(b) · 2022 Q24(c) · 2017 Q26 · 2025 Q2 · 2025 Q4(b)

Aldol condensation — ethanal to but-2-enal (dilute NaOH, then heat), and its cross version with benzaldehyde. The most repeated reaction in the chapter.

2019 Q9 · 2022 Q24(a) · 2023 Q31(b) · 2025 Q7(a)

Rosenmund reduction — an acid chloride with H₂/Pd–BaSO₄ gives the aldehyde (benzoyl chloride → benzaldehyde).

2022 Q11 · 2013 Q19(a) · Q29 · 2025 Q5(a)

Cannizzaro reaction — no α -hydrogen, concentrated alkali, one molecule oxidised and one reduced (methanal; benzaldehyde).

2020 Q23(a) · 2017 Q16(b) · 2025 Q7(b) · 2025 Q4(a)

Tollens' reagent — the silver-mirror test for an aldehyde against a ketone.

2017 Q12(b) · 2023 Q34(a) · 2025 Q7(b)

The sodium bicarbonate test for a carboxylic acid against an alcohol, phenol or ester.

2014 Q13(a) · 2022 Q15(a) · 2022 Q21(a) · 2022 Q27(a) · 2025 Q1

Reactivity in nucleophilic addition — aldehyde faster than ketone (sterics plus +I); benzaldehyde slower than propanal (resonance). Appears as a reason and as an ordering.

2022 Q5(a) · 2022 Q15(b) · 2022 Q21(b) · 2025 Q5(b)

Acid-strength explanations and orderings — electron-withdrawing groups stabilise the carboxylate; carboxylic acids beat alcohols by resonance.

2019 Q7 · 2022 Q24(b) · 2017 Q18(b)

Hell–Volhard–Zelinsky: $\text{Br}_2/\text{red P}$ (or Cl_2) puts a halogen on the α -carbon of the acid.

2017 Q12(a) · 2022 Q28(c) · 2023 Q35 · 2025 Q6(b)

Decarboxylation with soda lime (sodium salt of the acid $\rightarrow$ hydrocarbon, one carbon fewer; benzoate $\rightarrow$ benzene; ethanoate $\rightarrow$ methane).

2014 Q13(b) · 2022 Q27(c) · 2025 Q7(a)

Semicarbazone — the product, and why only one $-\text{NH}_2$ of semicarbazide reacts.

2014 Q13(a) · 2017 Q16(a) · 2020 Q23(b)

Cyanohydrin formation with HCN (CN^- as the nucleophile).

2022 Q28(a) · 2025 Q6(a) · 2025 Q7(a)

Acid chloride + dimethylcadmium gives a ketone (propanone, acetophenone).

2022 Q11 · 2022 Q28(b) · 2025 Q6(b)

Wolff–Kishner and Clemmensen — carbonyl to CH_2 (propanone $\rightarrow$ propane).

2017 Q18(a) · 2023 Q32 · 2025 Q3 · Q25

The iodoform test and the compounds that give it ($\text{CH}_3\text{CO}-$ or $\text{CH}_3\text{CH}(\text{OH})-$).

2017 Q6 · 2025 Q5(b)

Why carboxylic acids do not give the carbonyl reactions of aldehydes and ketones.

2022 Q14(a) · 2023 Q33(a)

The carboxyl group is deactivating and meta-directing; benzoic acid gives no Friedel–Crafts.

2017 Q2 · 2025 Q2

DIBAL-H reduces a nitrile to an aldehyde and leaves a $\text{C}=\text{C}$ alone.

2019 Q4 · 2025 Q4(a)

Ethylbenzene with KMnO_4 gives benzoic acid (via potassium benzoate in base).

Past year questions · numerical types — *no calculations in this chapter, so these are the reaction and structure types — one worked model each, then cold practice*

A · Multi-step conversion

Worked example · 2025 Q4(b) [2025] — benzaldehyde to benzophenone, and to 3-phenylpropan-1-ol

Benzophenone ($\text{C}_6\text{H}_5\text{COC}_6\text{H}_5$): oxidise the aldehyde to benzoic acid ($\text{K}_2\text{Cr}_2\text{O}_7$ in acetic acid), make the acid chloride (SOCl_2 , giving benzoyl chloride), then Friedel–Crafts acylation of benzene with AlCl_3 .

3-Phenylpropan-1-ol ($\text{C}_6\text{H}_5\text{CH}_2\text{CH}_2\text{CH}_2\text{OH}$): add two carbons by a cross aldol with ethanal (dilute NaOH) to $\text{C}_6\text{H}_5\text{CH}(\text{OH})\text{CH}_2\text{CHO}$; heat to lose water, giving cinnamaldehyde $\text{C}_6\text{H}_5\text{CH}=\text{CHCHO}$; then H_2/Pd reduces *both* the $\text{C}=\text{C}$ and the $-\text{CHO}$.

- 2017 Q18(a) [30+] — acetophenone to benzoic acid, in two steps — answer only: NaOH / I_2 (iodoform reaction) gives sodium benzoate + CHI_3 ; acidify
- 2017 Q18(b) [30+] — ethanoic acid to 2-hydroxyethanoic acid — answer only: $\text{Cl}_2 / \text{red P}$, then aqueous NaOH
- 2020 Q30 [30+] — benzene to 3-nitroacetophenone — answer only: $\text{CH}_3\text{COCl} / \text{AlCl}_3$ (acylation), then $\text{HNO}_3/\text{H}_2\text{SO}_4$ (the COCH_3 directs meta)
- 2020 Q30 [30+] — benzyl alcohol to phenylethanoic acid — answer only: SOCl_2 , then KCN , then hydrolysis
- 2022 Q24 [30+] — propan-1-ol to 2-bromopropanoic acid — answer only: $\text{K}_2\text{Cr}_2\text{O}_7 / \text{H}^+$ to propanoic acid, then $\text{Br}_2 / \text{red P}$

B · Identify the compounds

Worked example · 2023 Q32 [30+] — A is $\text{C}_3\text{H}_8\text{O}$; with Cu at 573 K it gives B; B does not reduce Fehling's but A's product gives a yellow precipitate C with I_2/NaOH

1. Copper at 573 K is dehydrogenation: A is a primary or secondary alcohol, and B is an aldehyde or ketone.
 2. B does not reduce Fehling's, so it is not an aldehyde: B is a **ketone**, and with three carbons it is **propanone**.
 3. A, the alcohol that gives a ketone, is the secondary alcohol **propan-2-ol**.
 4. Propanone has $\text{CH}_3\text{CO}-$, so with I_2/NaOH it gives the yellow precipitate of **iodoform** CHI_3 (C), plus sodium ethanoate.
- 2023 Q35 [30+] — A ($\text{C}_4\text{H}_8\text{O}_2$) hydrolyses to B and C — answer only: A ethyl ethanoate; B ethanoic acid; C ethanol
 - Q25 [30+] — alkene C_5H_{10} whose ozonolysis products are ethanal and propanone — answer only: A 2-methylbut-2-ene; B ethanal; C propanone
 - Q29 [30+] — two moles of A with a strong base give B and C; B with Cu regenerates A; acidifying C gives CH_2O_2 — answer only: A methanal; B methanol; C sodium formate; D formic acid (Cannizzaro)
 - 2025 Q3 [2025] — A is $\text{C}_8\text{H}_8\text{O}$, positive 2,4-DNP, no Tollens', iodoform positive; drastic oxidation gives C — answer only: A acetophenone (1-phenylethanone); with $\text{NaOI} \rightarrow$ sodium benzoate + CHI_3 ; C benzoic acid

- 2025 Q2 [2025] — A ($\text{C}_4\text{H}_5\text{N}$) with DIBAL-H then water gives B, which gives Tollens' but not iodoform, and is also made from ethanal and dilute NaOH on heating — answer only: A but-2-enenitrile $\text{CH}_3\text{CH}=\text{CHCN}$; B but-2-enal

C • Give a reason

Worked example • 2022 Q15(a) [30+] / 2025 Q1 [2025] — benzaldehyde is less reactive than propanal in nucleophilic addition

In benzaldehyde the carbonyl is conjugated with the ring, so the ring's π electrons are drawn towards the carbonyl by **resonance**. That **lowers the positive charge** on the carbonyl carbon (it becomes **less electrophilic**), so the nucleophile finds less to attack. Propanal has no resonance, so its carbon is more positive and it reacts faster.

- 2022 Q14(a) [30+] — benzoic acid gives no Friedel–Crafts — answer only: COOH deactivates the ring, and AlCl_3 bonds to the carboxyl oxygen
- 2022 Q15(b) [30+] — 4-nitrobenzoic acid is a stronger acid than benzoic acid — answer only: $-\text{NO}_2$ is electron-withdrawing and stabilises the conjugate base
- 2017 Q6 [30+] — carboxylic acids do not give the reactions of the carbonyl group — answer only: the lone pair on $-\text{OH}$ is in resonance with $\text{C}=\text{O}$, lowering the carbon's electrophilicity
- 2014 Q13(a) [30+] — CH_3CHO is more reactive than CH_3COCH_3 towards HCN — answer only: two methyls cause more steric hindrance and more +I effect than one
- 2023 Q34(b) [30+] — α -hydrogens of an aldehyde or ketone are acidic — answer only: the carbonyl is electron-withdrawing, and the carbanion is resonance-stabilised

D • Distinguish

Worked example • 2020 Q23(a) [30+] — distinguish propanal from propanone

Warm each with freshly prepared Tollens' reagent (ammoniacal silver nitrate). Propanal is oxidised to propanoate and the silver ion is reduced to a **bright silver mirror**; propanone gives **no reaction**.

- 2017 Q12(b) [30+] — benzoic acid from phenol — answer only: NaHCO_3 ; the acid gives brisk effervescence of CO_2 , phenol does not
- 2017 Q16(b) [30+] — butanal from butan-2-one — answer only: Tollens' reagent; butanal gives a silver mirror
- 2025 Q7(b) [2025] — ethyl benzoate from benzoic acid — answer only: NaHCO_3 ; only the acid gives effervescence

E • Arrange in order

Worked example • 2022 Q21(a) [30+] — nucleophilic-addition reactivity of ethanal, propanone, propanal, butanone

Fewer, smaller groups on the carbonyl means faster. Ethanal has one methyl; propanal one ethyl; propanone two methyls; butanone a methyl and an ethyl. Increasing reactivity: **butanone < propanone < propanal < ethanal**.

- 2022 Q5(a) [30+] — acid strength of CH_3COOH , ClCH_2COOH , FCH_2COOH — answer only: $\text{CH}_3\text{COOH} < \text{ClCH}_2\text{COOH} < \text{FCH}_2\text{COOH}$
- 2022 Q5(b) [30+] — boiling points of CH_3CHO , $\text{CH}_3\text{CH}_2\text{OH}$, $\text{CH}_3\text{CH}_2\text{CH}_3$ — answer only: $\text{CH}_3\text{CH}_2\text{CH}_3 < \text{CH}_3\text{CHO} < \text{CH}_3\text{CH}_2\text{OH}$

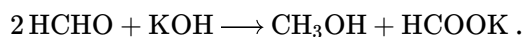
- 2022 Q21(b) [30+] — acid strength of 4-nitrobenzoic, benzoic, 3,4-dinitrobenzoic and 4-methoxybenzoic acids — answer only: 4-methoxy < benzoic < 4-nitro < 3,4-dinitro

F • Named reactions

Worked example • 2022 Q11 [30+] — Wolff–Kishner reduction and the Cannizzaro reaction

Wolff–Kishner: an aldehyde or ketone is heated with hydrazine to form the hydrazone (water lost), which is then heated with KOH in ethylene glycol; N₂ escapes and the carbonyl becomes CH₂:

$\text{CH}_3\text{COCH}_3 \longrightarrow \text{CH}_3\text{CH}_2\text{CH}_3$. **Cannizzaro:** an aldehyde with no α -hydrogen is heated with concentrated alkali; one molecule is reduced to the alcohol and one is oxidised to the carboxylate:



- 2015 Q22 [30+] — Stephen reaction — answer only: RCN with SnCl₂/HCl gives the imine, then H₃O⁺ gives RCHO
- 2015 Q22 [30+] — Etard reaction — answer only: toluene with CrO₂Cl₂ in CS₂ gives a chromium complex, hydrolysed to benzaldehyde
- 2025 Q5(a) [2025] — toluene with CrO₃ in acetic anhydride (A), hydrolysed (B), then concentrated NaOH (C) — answer only: A benzylidene diacetate; B benzaldehyde; C benzyl alcohol (with sodium benzoate); the video calls the first step Etard, but it is the CrO₃/Ac₂O route
- 2015 Q17 [30+] — acetylation — answer only: introducing CH₃CO– into an alcohol, phenol or amine with acetyl chloride or acetic anhydride (aniline gives acetanilide)

G • IUPAC names and structures

Worked example • 2014 Q1 [30+] — name the compound with a ketone at C2 and an alcohol at C4 of a five-carbon chain

The ketone outranks the alcohol, so it is the suffix and gets the lowest number (carbon 2); the alcohol becomes the prefix **hydroxy** at carbon 4. Five carbons, all single bonds: **4-hydroxypentan-2-one**.

- 2015 Q20 [30+] — structure of 4-methylpent-3-en-2-one — answer only: CH₃–CO–CH=C(CH₃)–CH₃
- 2015 Q20 [30+] — structure of *p*-methylbenzaldehyde — answer only: benzene ring with –CHO at C1 and –CH₃ at C4